University of Europe for Applied Sciences

RAG-based System for Composable Task Learning in Robotics

MASTER'S THESIS

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Problem Statement

01

The Intent Gap:

Disconnection between ambiguous natural language (e.g. "clean up") and the rigid , low-level code robots requirements (e.g. gripper.move_to(x,y,z)).

02

Safety & Hallucination Risks:

Standard Large Language Models (LLMs) lack a validation layer, often generating non-existent API commands or physically impossible trajectories.

03

High Barrier & Inflexibility:

Current automation relies on experts, changing a task requires expensive code rewriting rather than simple conversational adjustment.

Research Queations

01 To what extent does a Dual-RAG–based task grounding architecture improve the safety, structural validity, and execution reliability of robotic manipulation tasks compared to conventional LLM-based and single-RAG planning approaches in an industrial automation context?

02 Can the proposed conversational task planning and validation framework meet real-time performance requirements for robotic manipulation while maintaining execution accuracy in high-fidelity industrial simulation environments?

03 Measuring by tokens and time, how much enhancement is achieved via conversational refinement compared to full plan regeneration?

04 To what degree can a generative teach mode compilation approach reduce expert programming dependency by enabling non-expert operators to define new robotic manipulation tasks that are immediately executable and reusable in an industrial automation pipeline?

05 How effectively does a constraint-based conversational task adaptation mechanism enable runtime modification of robotic task plans while preserving safety constraints and minimizing re-planning overhead in dynamic manufacturing scenarios?

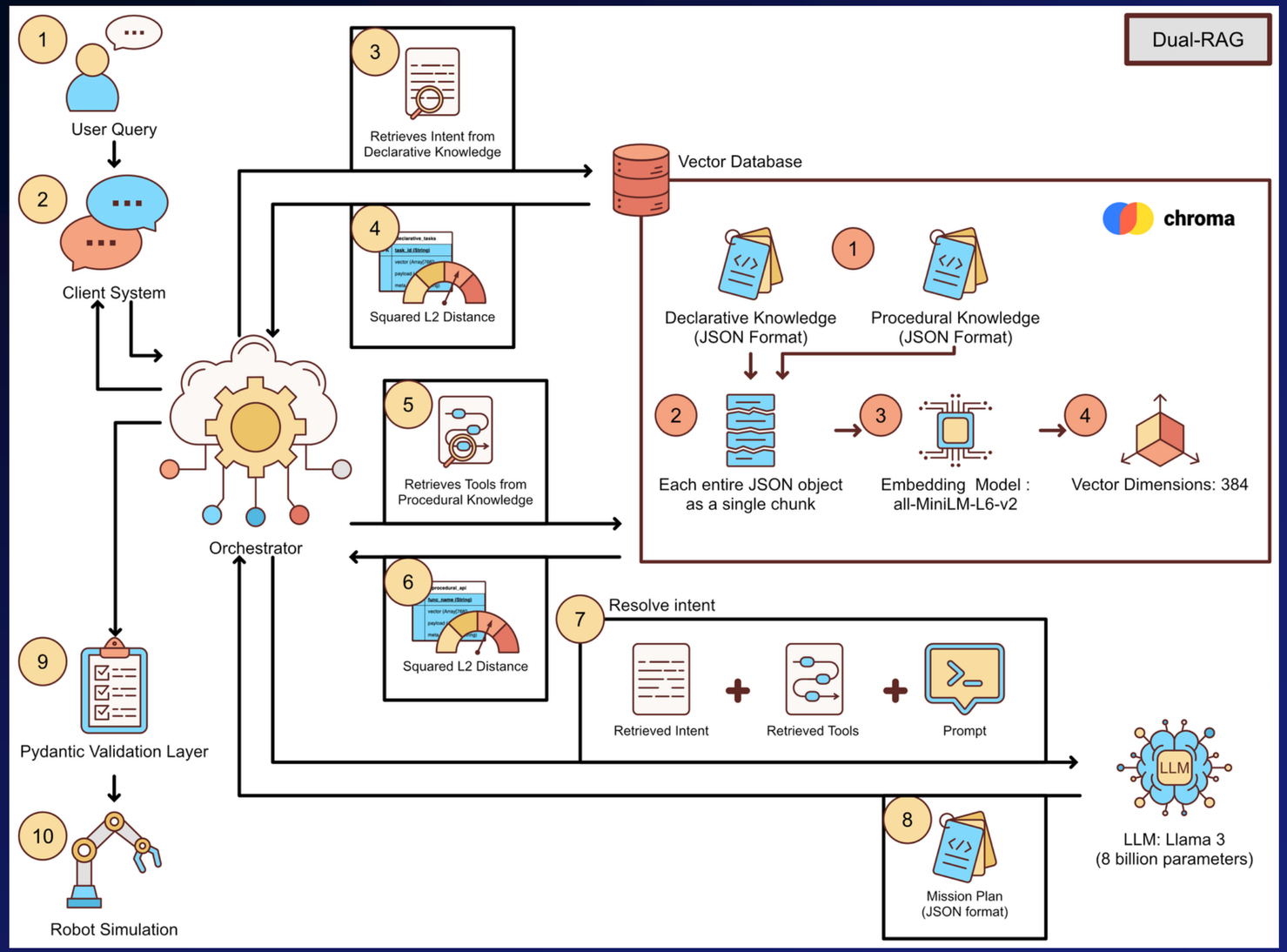
Literature Review

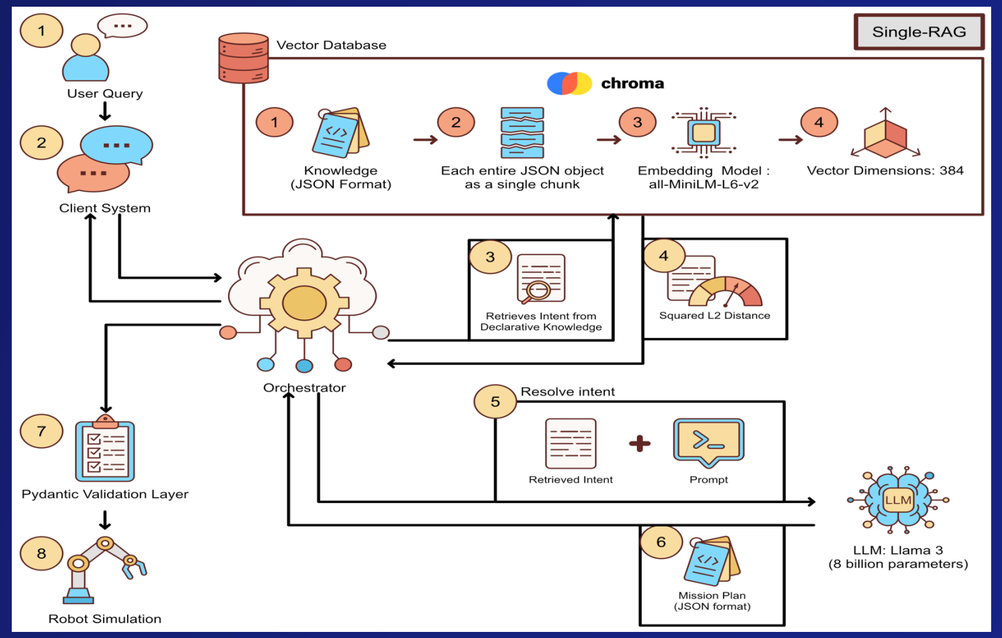
Topic	Reference	Methodology	Dataset	Environment	Challenges Addressed	Limitations
Model-Based Robotic Control with Physics-Based Simulation	Xu et al. (2026)\cite{XU2026103137}	Physics-based Hybrid Modeling	Machining Logs: Vibration, cutting force, and error logs from physical cutting tests.	CNC Machine Tool: Physical 5-axis milling machine instrumented with accelerometers.	Reducing physical vibration and error.	Focuses on low-level precision, not high-level planning.
	Wang et al. (2024)\cite{wang2024deep}	Digital Twin + Predictive DL	Hybrid Dataset: 20,823 synthetic images (from DT) and 13,926 real images (from physical cell) under 4 illumination levels.	Cobot Cell: Universal Robot 10 (UR10) monitored by Kinect V2, DT built in Unreal Engine 4.	Data scarcity in HRC; bridging "Sim-to-Real" gap for safety.	Accuracy depends entirely on simulation fidelity.
	Zhang et al. (2023)\cite{zhang2023deep}	Deep Learning and CP Fusion (PointNet and ICP)	Multimodal Sensor Data: Real-time RGB images combined with force sensor logs (Self-collected).	AR-Assisted Assembly Cell: Real robot synchronized with AR glasses and Digital Twin for human guidance.	Synchronizing real and virtual data for safety.	High latency in processing fused data.
Task-Level Programming via Symbolic Planning and Behavior Trees	Qi et al. (2026)\cite{qi2026llm}	Symbolic subgoal decomposition: via VLM query mechanism integrated with 3D Diffusion Policy.	Visual frames from human demonstrations and point cloud observations.	Xarm-7 manipulator with a customized pulley/idler task board.	Managing infinite-dimensional state spaces and long-horizon sequential dependencies of deformable object	Compounding errors and lack of hierarchy in traditional imitation learning.
	Tong et al. (2026)\cite{TONG2026103169}	GNN + LLM	Scene Graphs: Structured object relationship data (e.g., disassembly trees).	Collaborative Disassembly Cell	Reasoning about complex disassembly structures.	GNNs are computationally heavy to train or inference.
	Ma et al. (2023)\cite{MA2023102539}	TAMP + LfD	Task Trajectories: Library of "expert" trajectories recorded by humans moving the robot arm.	Robotic Manipulator	Adapting plans when packages change size or position.	"Cold start" problem, needs expert data first.

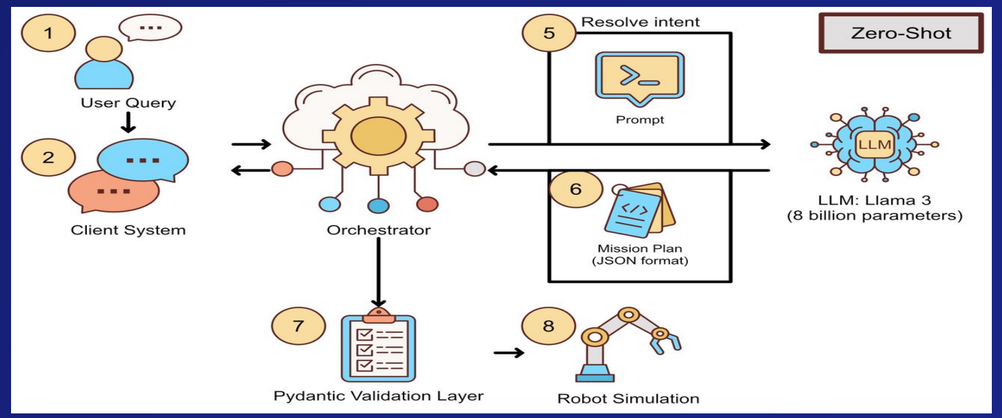
Literature Review

Topic	Reference	Methodology	Dataset	Environment	Challenges Addressed	Limitations
Learning-Based Control using Reinforcement Learning (RL)	Xue et al. (2026)\cite{XUE2026103159}	Hierarchical RL	Spatial Maps: Grid-based factory maps and task allocation logs.	Production Floor Simulation: 3D simulator of a complex real-time production process.	Allocating tasks to humans or robots while avoiding spatial bottlenecks.	"Black box" decision making; hard to explain why a path was chosen.
	Fan et al. (2026)\cite{FAN2026103215}	LLM-Guided RL	Simulation Episodes: Thousands of State-Action-Reward pairs generated during training.	Digital Twin Industrial Environment: Simulated factory floor used for rapid RL training episodes.	Improving the slow learning speed of RL.	Still requires significant training time; safety is probabilistic.
	Lin et al. (2025)\cite{2025five}	LRLDT: 5-Dimensional Digital Twin framework with RL	Tool-axis vector variation rates and collision interference metrics.	5-axis CNC machining for complex components.	Overcoming dependency on human expertise in tool-axis vector planning to prevent collisions.	LLM used only as an indirect decision-maker rather than for direct real-time control.
Natural Language - Driven Control using LLMs and Tool-Grounded Execution	Wang et al. (2026)\vite{wang2026llm}	HRC Agent: Chain-of-Thought (CoT) reasoning and reflection-based memory.	Visual feature data via YOLOv8 and natural language instructions from operators.	Aerospace wire harnessing assembly.	Programming complexity for varying batch sizes and customized production tasks.	Traditional LLM fine-tuning can damage inherent reasoning abilities and is computationally expensive.
	Shi et al. (2026)\cite{shi2026cognitive}	CDT and CoT-RAG: AAS and CoT-prompting for inference.	106 historical production orders (2021 - 2023) and reusable AAS submodel templates.	Machining Center: A small-scale production facility with 6 machines (2 CNC lathes, 4 milling machines).	Semantic Disconnect: Overcoming the gap between static product requirements and flexible machine capabilities; reducing reliance on manual configuration.	Textual descriptions in AAS submodels may not capture full workpiece geometry without multimodal input.
	Li et al. (2026)\cite{li2026 3dprintmind}	Dynamic MKG-RAG: RAG using TGN for time-aware reasoning.	Printing Logs & Ontology: G-code files and structured material properties database.	Additive Manufacturing Farm: 3D Printers networked to a monitoring server.	Preventing LLM hallucinations in technical printing parameters; handling dynamic time-series data.	High computational requirements for processing dynamic representations in real-time.
Natural language - enabled industrial robotics and digital twin interaction with large language models	Yu et al. (2026)\cite{YU2026103113}	Systematic Review	N/A	General Industrial Robotics: Covers logistics, assembly, and inspection.	Summarizing state-of-the-art and identifying future gaps.	Identifies gaps but offers no new solution.
	Chen et al. (2025)\cite{CHEN2025102982}	Interactive-DT: Conceptual framework for integrating LLMs with Digital Twins.	N/A	Industry 5.0 Context: General concept of human-centric sustainable manufacturing.	Managing data transmission and analytics in large scale DT systems.	Theoretical, lacks specific algorithm implementation.
	Fogli et al. (2022)\cite{FOGLI2022102234}	Hybrid Interface (User Study)	User Study Data: Task completion times and error rates from human participants.	Collaborative Robot (Cobot)	Usability for non-expert operators.	Results depend on the specific user group tested.

Methodology







RQ 1:

To what extent does a Dual-RAG–based task grounding architecture improve the safety, structural validity, and execution reliability of robotic manipulation tasks compared to conventional LLM-based and single-RAG planning approaches in an industrial automation context?

Structural Validity and Safety Comparison

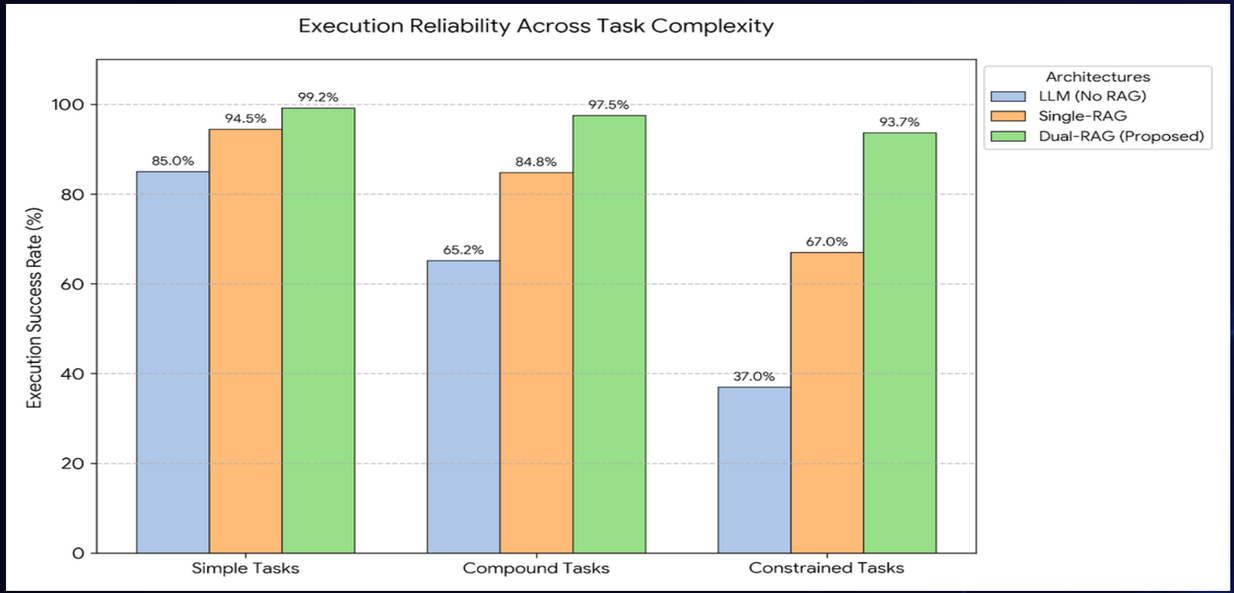
Comparison of structural validity, safety violations, and hallucination rates across different planning architectures.

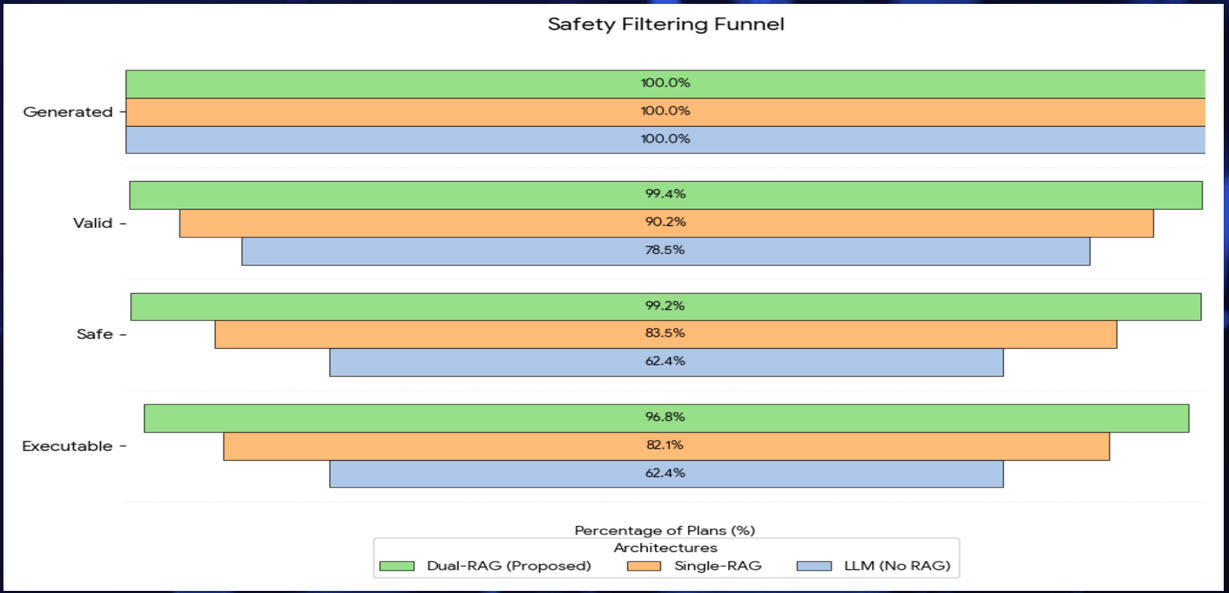
Architecture	Invalid JSON (%)	Safety Violation (%)	API Hallucination (%)	Execution Success (%)
LLM GPT-4 (No RAG)	21.5	16.2	18.9	62.4
Standard RAG	9.8	6.7	5.4	82.1
Dual-RAG (Proposed)	0.6	0.2	0.4	98.8

Failure Mode Distribution

Distribution of failure types observed during robotic task execution.

Architecture	Syntax Errors (%)	Safety Limit Violations (%)	Missing API Calls (%)	Logical Sequence Errors (%)
LLM GPT-4 (No RAG)	38	27	21	14
Standard RAG	19	11	8	6
Dual-RAG (Proposed)	2	1	1	1





RQ 2:

Can the proposed conversational task planning and validation framework meet real-time performance requirements for robotic manipulation while maintaining execution accuracy in high-fidelity industrial simulation environments?

End-to-End Latency Breakdown

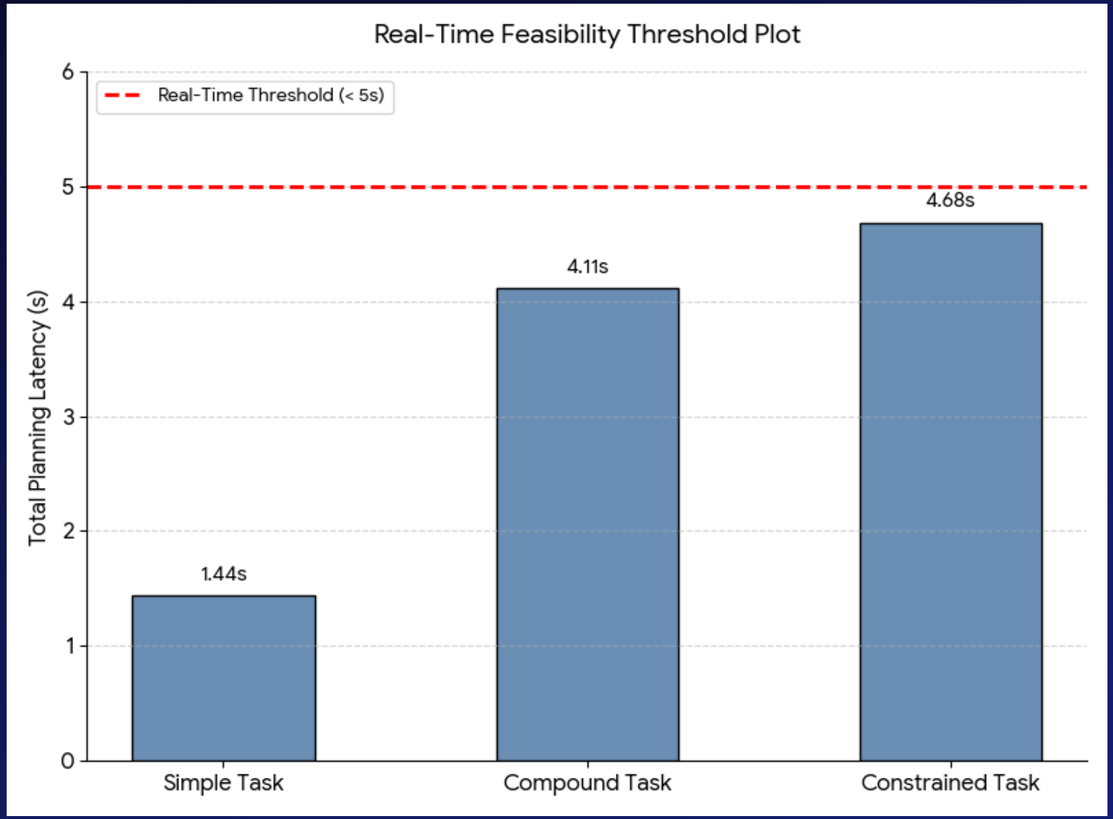
Latency contribution of each system component during robotic task planning.

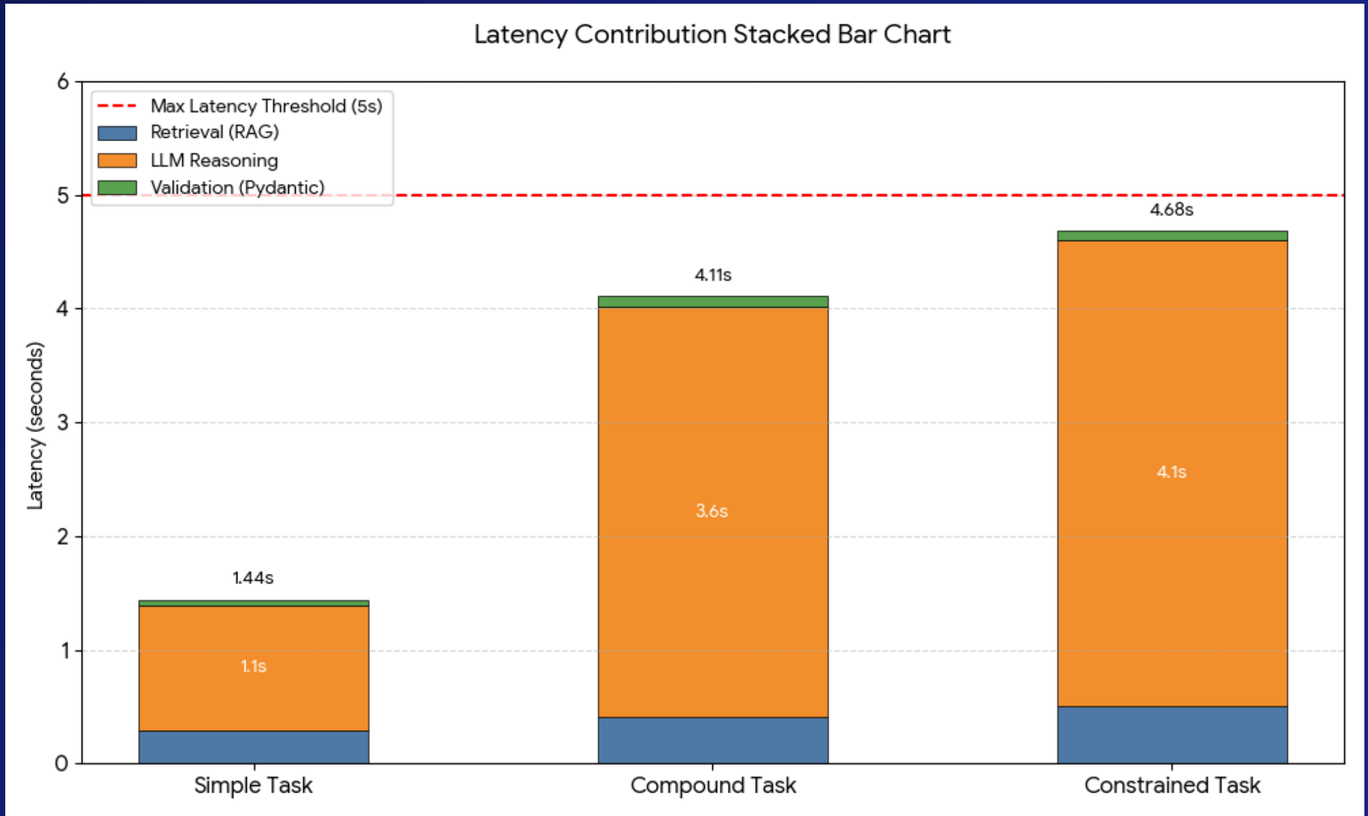
Task Category	Retrieval Time	LLM Generation Time	Validation Time	Total Latency
Simple Task (Single-step)	0.28s	1.10s	0.06s	1.44s
Compound tasks (Multi-step)	0.41s	3.62s	0.08s	4.11s
Constrained Task	0.52s	4.05s	0.11s	4.68s

Execution Accuracy in Simulation

Execution success and collision-free completion rates in high-fidelity simulation.

Task Category	Success Rate (%)	Collision-Free (%)	Kinematic Feasibility (%)
Simple Task (Single-step)	99.20	99.80	100.00
Compound tasks (Multi-step)	96.50	97.10	98.90
Constrained Task	93.40	95.00	96.20





RQ 3:

Measuring by tokens and time, how much enhancement is achieved via conversational refinement compared to full plan regeneration?

Token Consumption Comparison

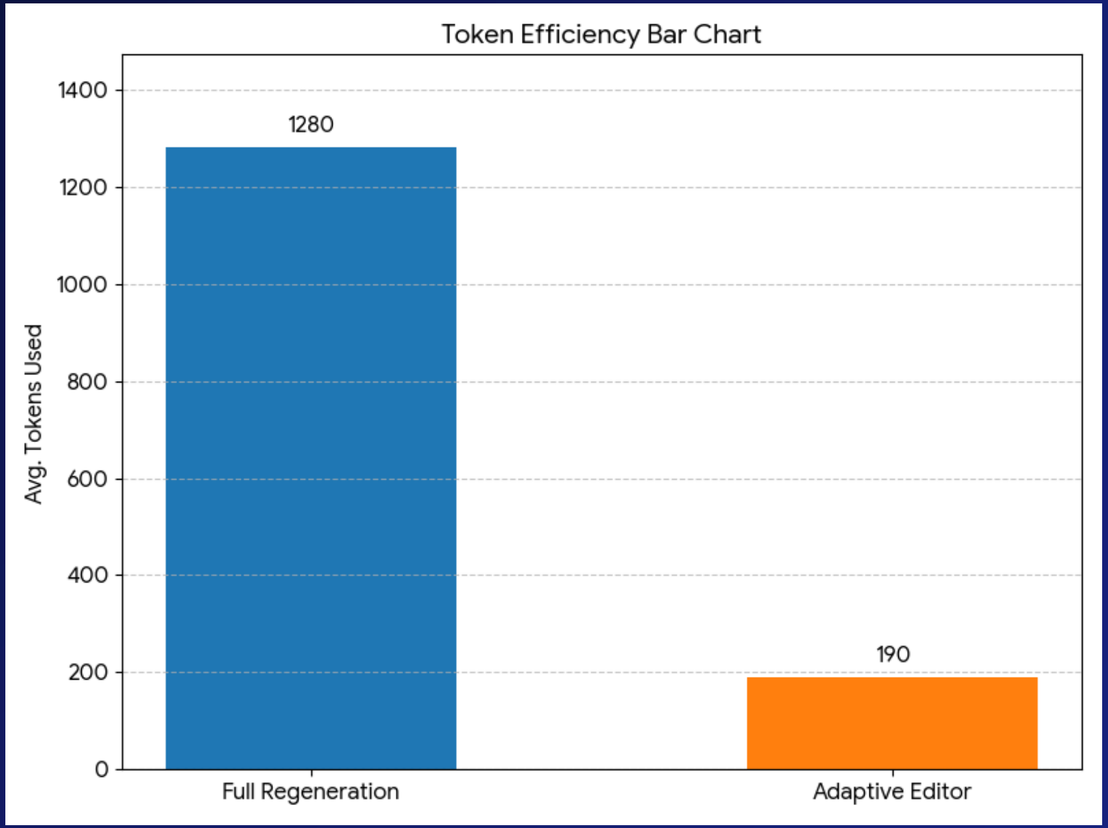
Token usage comparison between full plan regeneration and adaptive conversational editing.

Method	Avg. Tokens Used	Token Reduction (%)
Full Regeneration	1280	-
Adaptive Editor	190	85.2

Response Time Comparison

Average system response time for task modification.

Method	Avg. Response Time (s)
Full Regeneration	4.35
Adaptive Editor	0.92



RQ 4:

To what degree can a generative teach mode compilation approach reduce expert programming dependency by enabling non-expert operators to define new robotic manipulation tasks that are immediately executable and reusable in an industrial automation pipeline?

Task Creation Success Rate

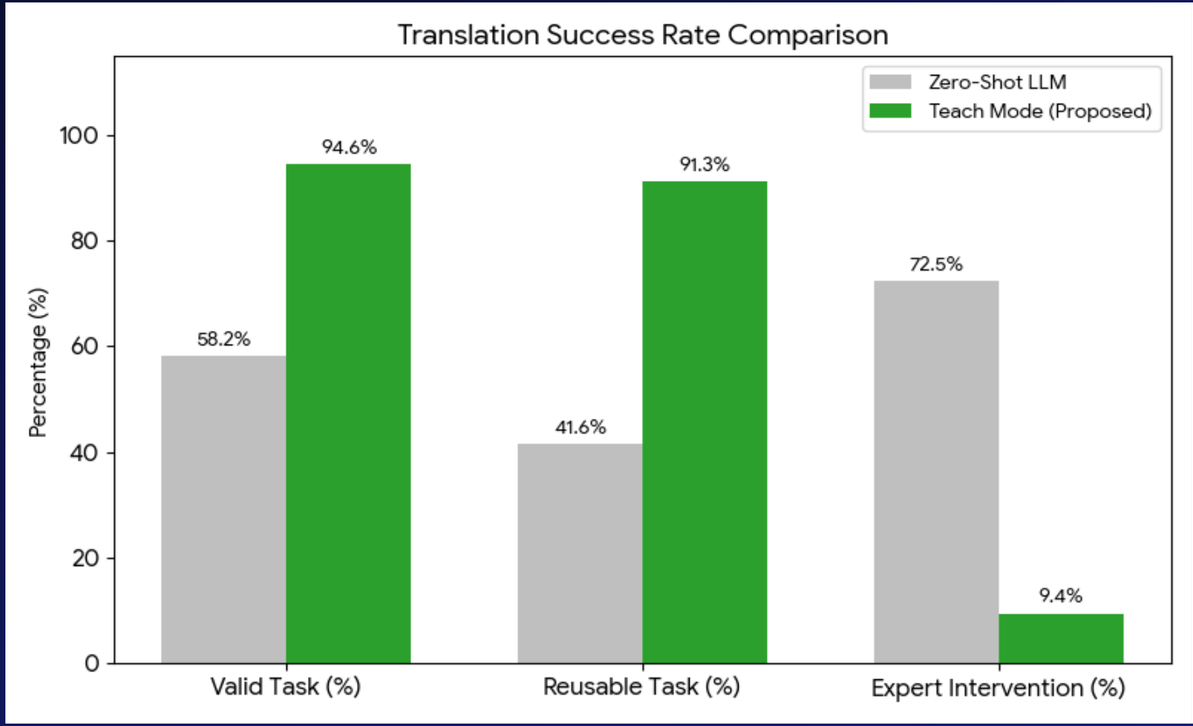
Comparison of task creation success between zero-shot LLMs and Generative Teach Mode.

Approach	Valid Task (%)	Reusable Task (%)	Expert Intervention Needed (%)
Zero-Shot LLM	58.2	41.6	72.5
Teach Mode (Proposed)	94.6	91.3	9.4

Error Distribution in Task Generation

Distribution of common errors in task generation.

Error Type	Zero-Shot LLM (%)	Teach Mode (%)
API Hallucination	34	3
Schema Violation	29	2
Logical Errors	21	4
Safety Violations	16	1



Zero-Shot LLM Output (Prone to Error)

```
def pick_obj():
  # Error: Hallucinated API
  robot.grab_object("red_box")

  # Error: Missing safety check
  robot.move_to(x=500, y=200)

Status: Failed (API Mismatch)
```

Teach-Mode Output (Proposed)

```
def pick_obj():
  # Validated API Call
  if robot.sense_force() < LIMIT:
    robot.gripper.close(force=0.5)

  # Safe Motion Planning
  robot.linear_move(target="bin_A")

Status: Success (Executable)
```

RQ 5:

How effectively does a constraint-based conversational task adaptation mechanism enable runtime modification of robotic task plans while preserving safety constraints and minimizing re-planning overhead in dynamic manufacturing scenarios?

Constraint Preservation Rate

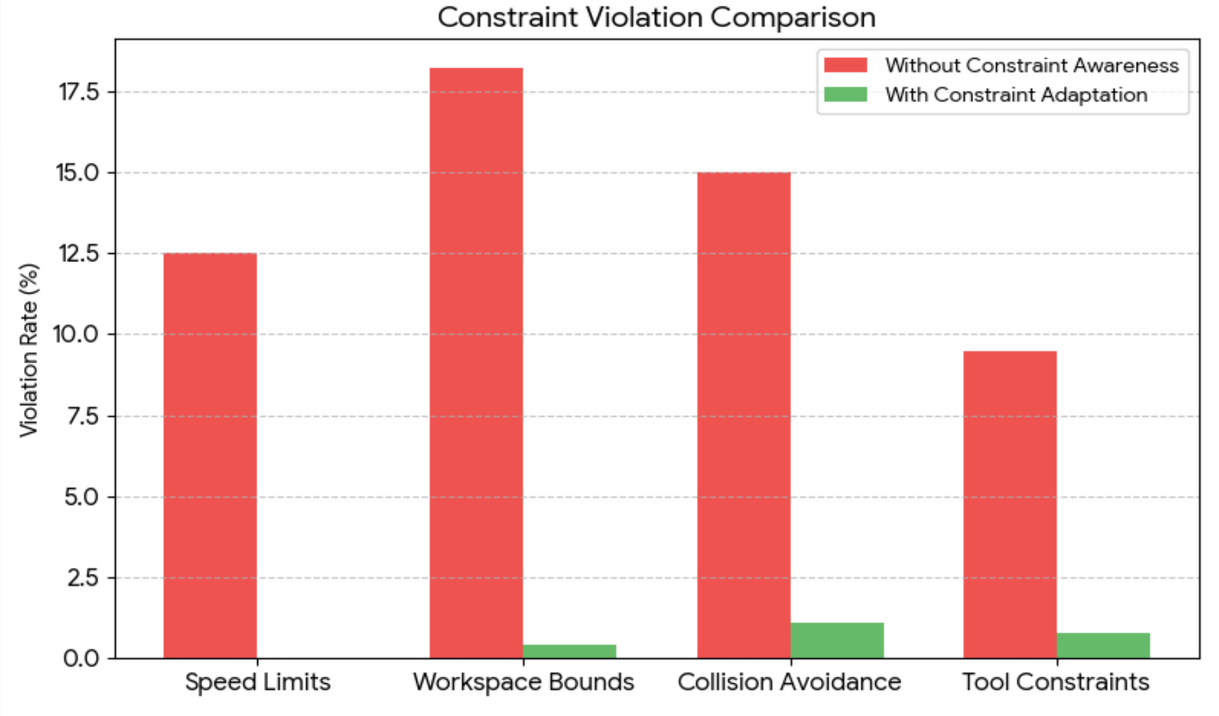
Constraint preservation accuracy during runtime task modification.

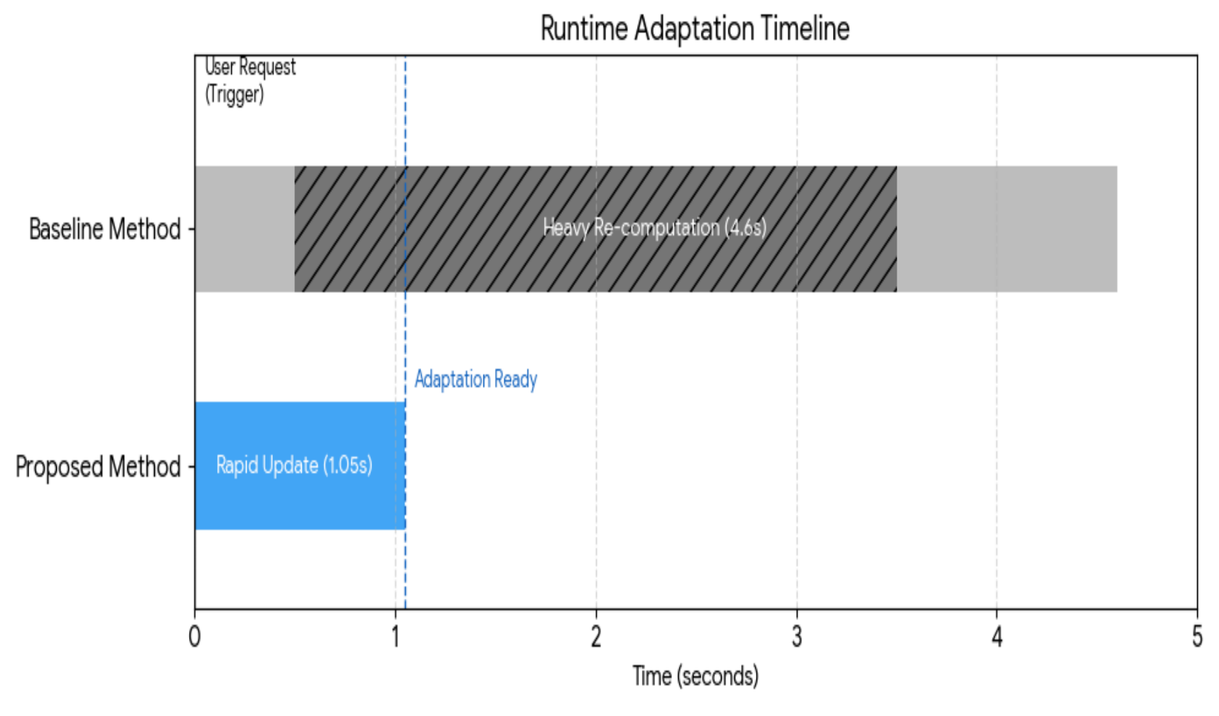
Constraint Type	Preservation Rate (%)
Speed Limits	100
Workspace Boundaries	99.6
Collision Avoidance	98.9
Tool Constraints	99.2

Re-planning Overhead Reduction

Caption: Reduction in computational overhead achieved through constraint-based adaptation.

Method	Avg. Re-planning Time (s)	Overhead Reduction (%)
Full Re-planning	4.6	-
Constraint-Based Adaptation	1.05	77.2





Conclusion

Contribution

- **Developed a Dual-RAG grounded architecture** that successfully bridges the gap between ambiguous human intent and rigid robotic code.
- **Implemented a three-tier agentic framework** (Retriever, Editor, Teacher) that separates high-level planning from low-level execution.

Key Findings

- Safety: Achieved 98.8% execution success rate, **effectively removing hallucinations via the procedural safety gate**.
- Efficiency: Adaptive Editor reduced computational overhead by 85.2%, **enabling cost-effective runtime modifications**.
- Democratization: Validated that **non-experts can define executable skills** with 94.6% success

Limitations

- **Simulation based:** Validation was conducted in a high-fidelity RoboDK digital twin, physical hardware dynamics (friction, sensor noise) were not tested.
- **Model Dependency:** Performance relies on the specific capabilities of the quantized Llama3.8b model.

Future Work

- Sim2Real Transfer: Deploying the architecture on a **physical Universal Robots** UR10e to validate robustness in the real world.
- Multimodal RAG: : **Incorporating visual feedback** to resolve spatial ambiguity.

Thank You

For Your Attention